Go2Silicon
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G2S6003

160Mhz 1.2V 0.13um Phase Lock Loop (PLL)

Description

G2S6003 is a general purpose clock generation PLL. The VCO frequency is optimized at 320Mhz for best jitter at this frequency. Input clock can be in the range of 5Mhz to 25Mhz depending on the application. Output frequency is 160Mhz. Also, a lock detector block is included for system monitoring purpose. Bias block and loop filter block are all integrated to provide a compact solution. This PLL has less than 20ps RMS jitter with a compact area of 0.12 mm^2. The design is in TSMC 0.13um 1.2V/3.3V process.

Features

- Uses single 1.2V power supply
- Fully integrated PLL including loop filter
- VCO operating frequency at 320Mhz
- Input frequency 5Mhz to 25Mhz
- Output frequency 160Mhz
- 0 ppm multiplication error
- 50% output clock duty cycle
- Optimized PLL loop bandwidth for best jitter performance
- RMS jitter less than 20ps
- With Lock indicator
- Fast lock time of less than 200us
- Low Power operation of only 8mA (26mW)
- Power down mode with reset control
- Compact silicon area of only 0.12mm^2
- TSMC 0.13um process

Applications

General purpose clock generation
Clock synthesizer
De-skew clock buffer